Incident-Guided Spatiotemporal Traffic Forecasting

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Abstract

Recent years have witnessed the rapid development of deep-learning-based, graph-neural-network-based forecasting methods for modern intelligent transportation systems. However, most existing work focuses exclusively on capturing spatio-temporal dependencies from historical traffic data, while overlooking the fact that suddenly occurring transportation incidents, such as traffic accidents and adverse weather, serve as external disturbances that can substantially alter temporal patterns. We argue that this issue has become a major obstacle to modeling the dynamics of traffic systems and improving prediction accuracy, but the unpredictability of incidents makes it difficult to observe patterns from historical sequences. To address these challenges, this paper proposes a novel framework named the Incident-Guided Spatiotemporal Graph Neural Network (**IGSTGNN**). IGSTGNN explicitly models the incident's impact through two core components: an Incident-Context Spatial Fusion (*ICSF*) module to capture the initial heterogeneous spatial influence, and a Temporal Incident Impact Decay (*TIID*) module to model the subsequent dynamic dissipation. To facilitate research on the spatio-temporal impact of incidents on traffic flow, a large-scale dataset is constructed and released, featuring incident records that are time-aligned with traffic time series. On this new benchmark, the proposed IGSTGNN framework is demonstrated to achieve state-of-the-art performance. Furthermore, the generalizability of the *ICSF* and *TIID* modules is validated by integrating them into various existing models.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; • **Information systems** → **Data mining**.

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Keywords

Traffic prediction, Incidents, Spatio-temporal data mining

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1 Introduction

In contemporary intelligent transportation systems (ITS), the accurate perception and prediction of urban traffic networks form a critical foundation for elevating intelligent city management to a new level [17, 42]. By leveraging large-scale historical data, traffic forecasting endeavors to uncover the underlying evolutionary dynamics of traffic flow, thereby enabling reliable predictions of future network states [5]. Such capability is indispensable for facilitating proactive traffic control, ensuring rapid emergency response, and enhancing the overall efficiency and resilience of urban road networks [44, 45].

In recent years, the dominant paradigm in this field has shifted towards data-driven approaches, particularly deep learning models designed to capture spatio-temporal dynamics. These models often abstract the traffic network as a graph structure, enabling the capture of complex spatial dependencies through graph-based neural network architectures [12, 31, 41]. For the temporal dimension, early models primarily combined Recurrent Neural Networks (RNNs) or Convolutional Neural Networks (CNNs) to model sequential dynamics [4, 6, 20, 30]. Furthermore, to capture more dynamic and long-range spatio-temporal correlations, many advanced models have incorporated attention mechanisms [33, 38]. These models excel at capturing the intrinsic recurrent patterns in traffic flow, such as the morning and evening peaks driven by commuting regularities, and have achieved state-of-the-art performance on numerous public benchmarks.

However, real-world traffic flow is far more complex than just regular patterns, as it is constantly affected by various external disturbances such as traffic accidents, road maintenance, or adverse weather [5, 9, 28]. These external disturbances are inherently unpredictable. In historical traffic datasets, periods of regular flow

are implicitly mixed with periods affected by these disturbances, often without explicit labels to differentiate them. When a model is trained solely on this mixed data, its optimization process forces it to learn a compromised function that represents an average of both normal and abnormal patterns, smoothing over the sharp dynamics caused by incidents [39]. This ultimately leads to significant performance degradation when the model encounters an acute incident. As illustrated in Figure 1, a model trained on normal scenarios learns the smooth, daily traffic flow pattern for a node n_1 (as shown in (a)). However, when an unforeseen collision occurs at a nearby intersection (as shown in (b)), the model's forecast can only follow this learned historical average. This renders it incapable of capturing the sharp, real-world drop in traffic flow, leading to a massive prediction error. Among these various disturbances, non-recurrent incidents are a primary and particularly challenging subclass, and thus form the focus of this paper.

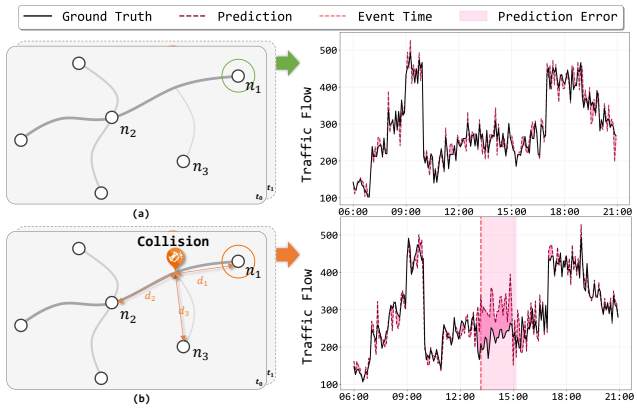


Figure 1: Illustration of a conventional model's failure under incident conditions. (a) Depicts the road network under a normal scenario, along with the typical daily traffic flow at node n_1. (b) Depicts the same network during an incident scenario with a collision at an intersection. The chart on the right focuses on the post-incident traffic flow at node n_1: the ground truth (black solid line) drops sharply, while the traditional model's forecast (dark red dashed line) completely misses this change. The pink vertical dashed line indicates the incident's start time, the pink vertical block highlights the two-hour impact duration, and the shaded red area quantifies the significant prediction error.

To effectively integrate information from such incidents and model their impact within traffic forecasting, three major challenges must be addressed. (1) **Non-uniform Incident Impact on Spatial Domain**. Incidents could occur at any location in the road network. Even if two incidents occur between the same pair of traffic sensors, the sensors may still perceive different impacts due to the varying distances between the accident points and the sensors [32, 40]. (2) **Dynamic Temporal Evolution of Impact**. The influence of an incident is not a singular, static impulse but a dynamic process that evolves over time [3]. Failing to model this temporal decay is a primary cause of severe long-term prediction errors. (3) **Diversity and Complexity of Incident Information**. The information associated with traffic incidents is characterized by both high diversity and complexity. Incidents are diverse in their nature (e.g., a multi-vehicle collision versus a stalled car), and each type induces a unique impact on traffic flow. Compounding this, the data describing any single incident is itself complex, often presented in multi-modal and unstructured formats.

To address the aforementioned challenges, we propose a novel framework named the Incident-Guided Spatiotemporal Graph Neural Network (IGSTGNN). Our framework builds upon a traditional spatio-temporal modeling backbone by introducing two core modules to solve the problem of traffic forecasting under the influence of disruptions caused by non-recurrent incidents: an Incident-Context Spatial Fusion (*ICSF*) Module and a Temporal Incident Impact Decay (*TIID*) Module. First, the model encodes the various inputs, including: the traffic time series, the contextual features of incidents, and the static meta-features of the sensors. Next, to achieve a fine-grained spatial fusion of the disruption, the *ICSF* module employs an attention mechanism that dynamically evaluates the relationships between the incident, the regional characteristics of its location, and the real-time traffic state, generating a unique, incident-aware initial state representation for each network node. Subsequently, a spatio-temporal modeling module processes this incident-aware representation to simulate its propagation through the network. Finally, to model the dynamic temporal evolution of the incident, the *TIID* module explicitly models the physical process of the incident's diminishing impact over time and superimposes it onto the base traffic evolution trend, thereby generating more accurate and reliable long-term forecasts. The main contributions of this paper are three-fold:

- A novel incident-guided spatio-temporal forecasting framework, IGSTGNN, is proposed. Instead of solely learning from historical traffic patterns, this framework is designed to explicitly model the distinct spatio-temporal dynamics induced by external incidents.
- Two core components, *ICSF* and *TIID*, are proposed to model the incident's impact from both spatial and temporal dimensions. The *ICSF* module is responsible for capturing the initial, non-uniform spatial influence, while the *TIID* module models the subsequent dynamic dissipation of this impact over time.
- A comprehensive, real-world dataset for incident-guided forecasting is constructed and released to the public. On this benchmark, the proposed IGSTGNN framework is shown to achieve state-of-the-art performance. Moreover, the generalizability and effectiveness of the *ICSF* and *TIID* modules are further demonstrated by the improvement of integrating them into various existing models.

2 Preliminaries

In this section, we formally define the fundamental data structures and formulate the task of incident-aware traffic forecasting.

Definition 1. Traffic Network and Spatio-Temporal Time Series. The road network is represented as a weighted undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{v_1, \ldots, v_N\}$ is the set of N traffic sensors and $\mathcal{E}$ is the set of edges representing road connections. The spatial relationships between sensor nodes are described by

the weighted adjacency matrix $\boldsymbol{A} \in \mathbb{R}^{N\times N}$. The dynamic traffic conditions on this network are represented by a feature tensor $\boldsymbol{X} \in \mathbb{R}^{T\times N\times C}$, where a slice $\boldsymbol{x}_t \in \mathbb{R}^{N\times C}$ captures the state of all N sensors at time step t.

Definition 2. Contextual Features. For incident-aware forecasting, we incorporate two heterogeneous sources of contextual information: static sensor attributes and dynamic incident features. For each sensor $v_i \in \mathcal{V}$, its features are divided into two parts: semantic attributes $\boldsymbol{s}_{i,f}$ that directly describe the road segment for feature encoding (e.g., road type, lane width) and spatial attributes $\boldsymbol{s}_{i,s}$ that are used to determine spatial relationships and construct the network topology (e.g., latitude, longitude, Fwy, Abs PM). The raw features for all sensors are consolidated in a matrix $\boldsymbol{S}_{\text{raw}}$. For each dynamic incident e_k, its features are subdivided into three parts: (1) intrinsic attributes $\boldsymbol{e}_{k,\text{i}}$ that describe the event itself (e.g., incident type and description); (2) auxiliary contextual attributes $\boldsymbol{e}_{k,\text{c}}$ (e.g., the incident's relative position in the input sequence and holiday status); and (3) spatial attributes $\boldsymbol{e}_{k,\text{s}}$ used for spatial anchoring (e.g., latitude, longitude, Fwy, and Abs PM). The raw features for all incidents, encompassing these three categories, are consolidated in the matrix $\boldsymbol{E}_{\text{raw}}$.

Definition 3. Incident-Sensor Spatial Relationship Tensor. The spatial relationship between incidents and sensors is captured in a pre-defined tensor $\boldsymbol{D} \in \mathbb{R}^{M\times N\times 3}$. To comprehensively represent this relationship, we model it from three complementary perspectives. First, we consider two distinct distance metrics: **road network distance** is used to capture the primary, linear propagation of impact along direct road connections, while **Euclidean distance** models the secondary, "spillover" effect on geographically proximate areas (e.g., adjacent surface streets). Second, because an incident's impact is highly asymmetric and direction-dependent, the **upstream/downstream relationship** is included as a critical feature to differentiate the severe effects on upstream traffic from the often negligible effects on downstream traffic. For each incident-sensor pair (e_k, v_j), a corresponding 3-dimensional vector $\boldsymbol{D}_{kj,:}$ is constructed. Its first two components are proximity scores derived by applying a Gaussian kernel to the **Euclidean** and **road network distances**, respectively, while the third component is a binary indicator of their relative position (**1** if the incident's Absolute Postmile is greater, **0** otherwise). The tensor $\boldsymbol{D}$ thus encapsulates the multi-faceted and direction-dependent spatial relationship between incidents and sensors.

Definition 4. Problem Statement. The task is formulated based on a key assumption: the impact of incidents that occurred long ago is already implicitly captured within the recent historical traffic data. However, an incident that occurs at the most recent time step, t, presents a unique challenge, as its full impact has not yet propagated through the network and is not reflected in the historical sequence. Therefore, our task specifically focuses on predicting future traffic states by explicitly modeling the impact of these newly occurring incidents. Formally, let $\boldsymbol{X} = (\boldsymbol{x}_{t-T_h+1}, \ldots, \boldsymbol{x}_t)$ denote the historical traffic states and $\hat{\boldsymbol{Y}} = (\hat{\boldsymbol{y}}_{t+1}, \ldots, \hat{\boldsymbol{y}}_{t+T_p})$ denote the target prediction sequence. Given an input instance comprising the set $\{\boldsymbol{X}, \boldsymbol{S}_{\text{raw}}, \boldsymbol{E}_{\text{raw}}, \boldsymbol{D}\}$, which includes the historical traffic states, static sensor attributes, and information for new incidents occurring at time t, the objective is to predict the future traffic states $\hat{\boldsymbol{Y}}$ for the next T_p time steps. This task can be formulated as learning a mapping function f:

$$\hat{\boldsymbol{Y}} = f(\boldsymbol{X}, \boldsymbol{S}_{\text{raw}}, \boldsymbol{E}_{\text{raw}}, \boldsymbol{D}; \mathcal{G}, \Theta), \tag{1}$$

where Θ represents the learnable parameters of the model.

3 Methodology

Our proposed framework, named the Incident-Guided Spatiotemporal Graph Neural Network (IGSTGNN), is designed to accurately forecast traffic conditions by explicitly modeling the influence of non-recurrent incidents. As illustrated in Figure 2, the architecture follows a three-stage pipeline: (1) an Incident-Context Spatial Fusion (*ICSF*) Module to inject heterogeneous contextual information into the traffic representation; (2) a spatio-temporal (ST) modeling module to capture the propagation and evolution of incident shocks within the network; and (3) a Temporal Incident Impact Decay (*TIID*) Module to model the long-term decaying effect of the incident's impact and generate the final forecast.

3.1 Feature Encoder

The semantic attributes of the sensors ($\boldsymbol{s}_{i,f}$) provide rich contextual information about the road network, which is highly correlated with incident occurrence and impact [14]. Furthermore, an incident's intrinsic ($\boldsymbol{e}_{k,i}$) and contextual ($\boldsymbol{e}_{k,c}$) attributes are indispensable for the model to perceive the specific nature of each disruption.

Let $\boldsymbol{S}_f$ and $\boldsymbol{E}_{ic}$ denote the matrices consolidating these specific non-spatial features for all sensors and incidents, respectively. We design dedicated encoding functions, ϕ_S and ϕ_I, to transform these feature matrices into dense vector representations:

$$\boldsymbol{S} = \phi_S(\boldsymbol{S}_f), \quad \boldsymbol{I} = \phi_I(\boldsymbol{E}_{ic}), \tag{2}$$

where $\boldsymbol{S} \in \mathbb{R}^{N\times d_s}$ and $\boldsymbol{I} \in \mathbb{R}^{M\times d_e}$ are the resulting encoded representations for the sensors and incidents, respectively. Concurrently, the raw traffic time series $\boldsymbol{X}$ is also projected into a high-dimensional hidden space via a linear layer to obtain its initial hidden state representation $\boldsymbol{H} \in \mathbb{R}^{T_h\times N\times d_h}$.

3.2 Incident-Context Spatial Fusion

An incident's impact is not uniform across the network since it is significantly dependent on factors like the distance from the event and the characteristics of the affected area. To account for this spatially heterogeneous impact, the module quantifies the influence of each incident on every sensor, thereby creating a traffic representation that is contextually aware of the event.

As depicted in Figure 2(b), the *ICSF* module is designed to capture the initial, non-uniform spatial impact of an incident. By fusing the encoded incident features ($\boldsymbol{I} \in \mathbb{R}^{M\times d_e}$) and sensor attributes ($\boldsymbol{S} \in \mathbb{R}^{N\times d_s}$) with the most recent traffic state ($\boldsymbol{H}_t \in \mathbb{R}^{N\times d_h}$), guided by the pre-defined spatial relationship tensor ($\boldsymbol{D} \in \mathbb{R}^{M\times N\times 3}$), the module produces an updated hidden state representation, $\boldsymbol{H}'_t$, which is now conditioned on the incident's context.

Attention Component Projection. The core of the fusion process is a scaled dot-product attention mechanism. To facilitate this, we linearly project the traffic hidden states $\boldsymbol{H}_t$ and the encoded incident representations $\boldsymbol{I}$ into three distinct subspaces:

$$\boldsymbol{Q} = \boldsymbol{H}_t\boldsymbol{W}_Q, \quad \boldsymbol{K} = \boldsymbol{I}\boldsymbol{W}_K, \quad \boldsymbol{V} = \boldsymbol{I}\boldsymbol{W}_V, \tag{3}$$

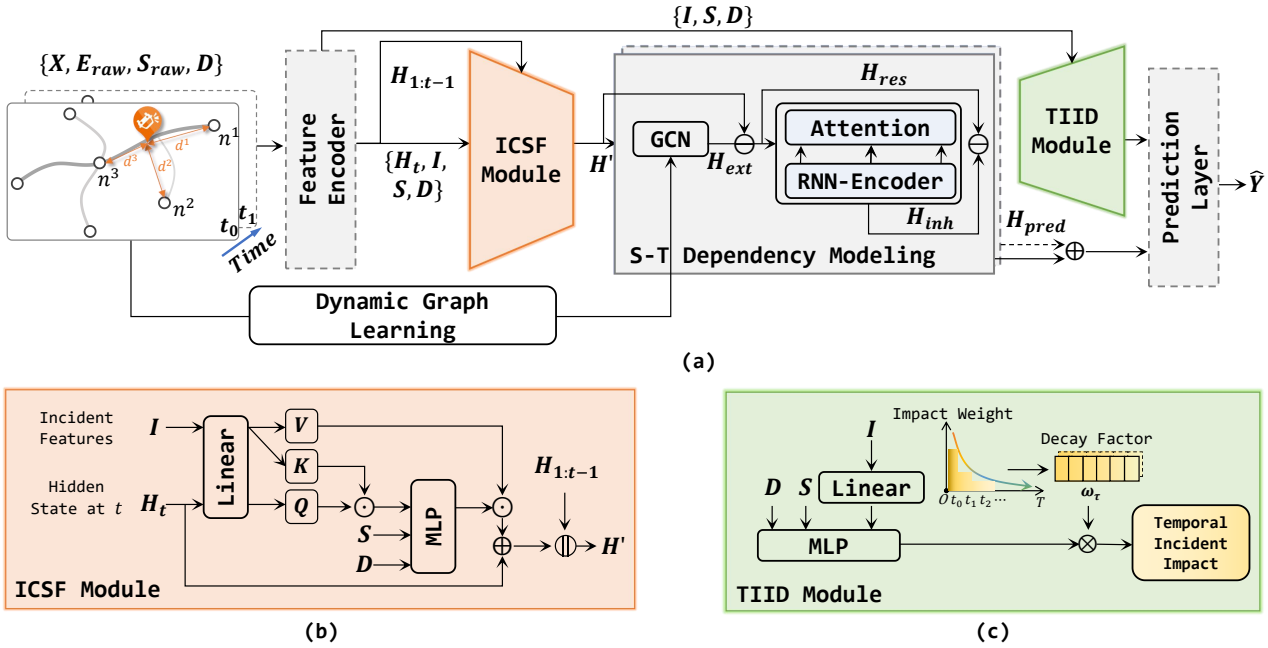


Figure 2: The overall architecture of our proposed IGSTGNN framework. (a) illustrates the main pipeline of the model, while (b) and (c) provide detailed views of the *ICSF* and *TIID* modules, respectively.

where $\boldsymbol{W}_Q \in \mathbb{R}^{d_h \times d_k}$, $\boldsymbol{W}_K \in \mathbb{R}^{d_e \times d_k}$, and $\boldsymbol{W}_V \in \mathbb{R}^{d_e \times d_v}$ are learnable projection matrices. The resulting $\boldsymbol{Q} \in \mathbb{R}^{N \times d_k}$ represents the current state of each sensor, $\boldsymbol{K} \in \mathbb{R}^{M \times d_k}$ represents the properties of the incidents for matching, and $\boldsymbol{V} \in \mathbb{R}^{M \times d_v}$ contains the rich information to be aggregated from the incidents.

Contextual Information Fusion via Attention. The central goal of this process is to compute a final attention weight matrix $\boldsymbol{\alpha}$ that reflects not only the interplay between traffic states and incident semantics but also incorporates prior knowledge from the road network topology.

First, we compute the initial semantic relevance score matrix via scaled dot-product attention:

$$\boldsymbol{A}_{\text{sem}} = \frac{\boldsymbol{K}\boldsymbol{Q}^T}{\sqrt{d_k}}, \tag{4}$$

where $\boldsymbol{A}_{\text{sem}}$ is the resulting score matrix, and each element $(\boldsymbol{A}_{\text{sem}})_{kj}$ quantifies the semantic affinity between the k-th incident and the hidden state of the j-th sensor.

However, $\boldsymbol{A}_{\text{sem}}$ is based purely on feature space similarity and does not account for topological constraints of the road network. An incident should not directly influence a distant node with which it has no topological connection. To enforce this strong prior, we define a masking function, Mask$(\cdot)$, based on the spatial relationship tensor $\boldsymbol{D}$. It operates on each element of $\boldsymbol{A}_{\text{sem}}$ as follows:

$$(\boldsymbol{A}'_{\text{sem}})_{kj} = \begin{cases} (\boldsymbol{A}_{\text{sem}})_{kj} & \text{if } (e_k, v_j) \text{ are connected in } \boldsymbol{D} \\ -\infty & \text{otherwise} \end{cases}. \tag{5}$$

Subsequently, the masked score matrix, $\boldsymbol{A}'_{\text{sem}}$, is normalized via the softmax function over the incident dimension for each sensor node to obtain the preliminary attention weights:

$$\tilde{\boldsymbol{\alpha}} = \text{softmax}(\boldsymbol{A}'_{\text{sem}}), \tag{6}$$

where $\tilde{\boldsymbol{\alpha}}$ is the resulting preliminary, spatially-aware weights.

While the preliminary weights, $\tilde{\boldsymbol{\alpha}}$, incorporate topological information, they do not yet consider the unique characteristics of each sensor's location. Therefore, the preliminary weights $\tilde{\boldsymbol{\alpha}}$, the broadcasted sensor features $\boldsymbol{S}$, and the spatial relationship tensor $\boldsymbol{D}$ are concatenated and processed by a fusion function g_α and a final softmax layer (also applied over the incident dimension) to generate the final attention scores:

$$\boldsymbol{\alpha} = \text{softmax}(g_\alpha(\tilde{\boldsymbol{\alpha}} \parallel \boldsymbol{S} \parallel \boldsymbol{D})), \tag{7}$$

where $\boldsymbol{\alpha}$ are the final context-aware attention weights, g_α is a fusion function implemented as an MLP, and $\parallel$ denotes the concatenation operation (with inputs broadcasted to compatible shapes).

Weighted Aggregation. The final attention weights $\boldsymbol{\alpha}$ are used to perform a weighted sum over the incident value representations $\boldsymbol{V}$, creating the incident context vector $\boldsymbol{C}$ for all sensor nodes:

$$\boldsymbol{C} = \boldsymbol{\alpha}^\top \boldsymbol{V}, \tag{8}$$

where $\boldsymbol{C} \in \mathbb{R}^{N \times d_v}$ condenses the most relevant incident information for each sensor.

Residual Fusion. Finally, the computed incident context $\boldsymbol{C}$ is added to the traffic hidden state $\boldsymbol{H}_t$ via a residual connection, and the result is normalized to produce the updated representation $\boldsymbol{H}'_t$:

$$\boldsymbol{H}'_t = \text{LayerNorm}(\boldsymbol{H}_t + \boldsymbol{C}), \tag{9}$$

where $\boldsymbol{H}'_t$ serves as the final, incident-aware hidden state for the subsequent spatio-temporal modeling blocks.

3.3 Spatio-Temporal Dependency Modeling

After the *ICSF* module injects the initial impact of an incident at the final time step, the ST modeling module is tasked with simulating how this disruption propagates through the network over space and time. To effectively model this complex process, our ST module adopts a decoupling principle, aiming to separately capture two distinct traffic patterns: the external influence propagated from neighboring nodes, and the node's own inherent temporal evolution. The module processes the incident-infused sequence $\boldsymbol{H}' \in \mathbb{R}^{T_h \times N \times d_h}$ iteratively through stacked decoupling blocks.

At the core of each decoupling block is a multi-graph convolution operation, designed to capture rich spatial dependencies from heterogeneous adjacency relationships. We leverage three distinct adjacency matrices simultaneously: (1) The pre-defined static adjacency matrix $\boldsymbol{A}$, which describes the fixed physical road network structure; (2) An adaptive adjacency matrix $\boldsymbol{A}_{\text{ada}}$, constructed from learnable node embeddings to capture latent static spatial dependencies as

$$\boldsymbol{A}_{\text{ada}} = \text{softmax}(\text{ReLU}(\boldsymbol{E}_u \boldsymbol{E}_d^T)), \tag{10}$$

where $\boldsymbol{E}_u, \boldsymbol{E}_d \in \mathbb{R}^{N \times d_{emb}}$ denote the learnable node embedding matrices; and (3) A dynamic adjacency matrix $\boldsymbol{A}_{\text{dyn}}$ [30], generated to capture transient spatial correlations that evolve over time:

$$\boldsymbol{A}_{\text{dyn}} = \text{softmax}\left(\frac{(\boldsymbol{E}_{\text{dyn}} \boldsymbol{W}_Q^{\text{dyn}})(\boldsymbol{E}_{\text{dyn}} \boldsymbol{W}_K^{\text{dyn}})^T}{\sqrt{d_{\text{dyn}}}} \right), \tag{11}$$

where $\boldsymbol{E}_{\text{dyn}} \in \mathbb{R}^{N \times d_{\text{dyn}}}$ represents the dynamic features (a concatenation of current traffic states and temporal embeddings), and $\boldsymbol{W}_Q^{\text{dyn}}, \boldsymbol{W}_K^{\text{dyn}} \in \mathbb{R}^{d_{\text{dyn}} \times d_{\text{dyn}}}$ are learnable projection matrices specific to this module.

In the l-th decoupling block, the input $\boldsymbol{H}^{(l-1)}$ is first processed by an **External Influence Component** using multi-graph convolution (g_S) to capture propagated spatial influences, yielding $\boldsymbol{H}_{\text{ext}}^{(l)}$. The inherent information flow of the nodes is then isolated via a residual connection:

$$\boldsymbol{H}_{\text{res}}^{(l)} = \boldsymbol{H}^{(l-1)} - \boldsymbol{H}_{\text{ext}}^{(l)}, \tag{12}$$

where $\boldsymbol{H}_{\text{res}}^{(l)}$ represents the residual component of the hidden state, intended to capture each node's inherent trend.

Next, an **Inherent Trend Component** employs a combination of a Recurrent Neural Network (RNN) and self-attention to process the residual representation $\boldsymbol{H}_{\text{res}}^{(l)}$ and produce the refined trend component $\boldsymbol{H}_{\text{inh}}^{(l)}$. To obtain the final output of the l-th block, $\boldsymbol{H}^{(l)}$, this refined trend is then subtracted from the original residual, $\boldsymbol{H}_{\text{res}}^{(l)}$, in a manner analogous to the residual calculation in Equation (12). This final output, $\boldsymbol{H}^{(l)}$, subsequently serves as the input for the $(l+1)$-th layer.

After L iterations, we accumulate the forecast parts generated by the two components from each layer to obtain the final Modeled Hidden State, $\boldsymbol{H}_{\text{pred}}$. This output is a high-level representation over the prediction horizon that captures the incident-conditioned spatio-temporal dynamics and provides a robust foundation for the subsequent *TIID* module to refine predictions via explicit decay modeling.

3.4 Temporal Incident Impact Decay

A key characteristic of traffic incidents is the dynamic nature of their impact, which evolves over time and naturally dissipates as the network recovers. Conventional models often fail to capture this temporal evolution. Therefore, our *TIID* module is specifically introduced to model this decay process, simulating the dissipation of an incident's influence throughout the forecast horizon.

As shown in Figure 2(c), the module receives the base forecast's hidden states $\boldsymbol{H}_{\text{pred}} \in \mathbb{R}^{T_p \times N \times d_{out}}$ from the ST module. To model the incident's decay effect, the module first computes the incident's initial spatial impact at the moment of occurrence, time t, which we term the initial incident context $\boldsymbol{C}_{\text{init}}$.

This computation aims to distill the potential impact of the incident on each node. Accordingly, the incident Key representation $\boldsymbol{K}$ is first expanded and spatially masked by the relationship tensor $\boldsymbol{D}$ to produce a localized incident representation, $\boldsymbol{K}_{\text{context}}$. This step ensures an incident only affects its spatially connected nodes. This localized representation is then concatenated with the sensor features $\boldsymbol{S}$ (dimensionally aligned via broadcasting) and the spatial relationship tensor $\boldsymbol{D}$ itself. Finally, this fused tensor is processed by a function g_c to generate the initial incident context:

$$\boldsymbol{C}_{\text{init}} = g_c(\boldsymbol{K}_{\text{context}} \parallel \boldsymbol{S} \parallel \boldsymbol{D}), \tag{13}$$

where g_c is a fusion function implemented as an MLP and $\parallel$ denotes the concatenation operation.

The *TIID* module employs a temporal decay factor ω_τ, which is modeled by a Gaussian function [13]. For each future prediction step $\tau \in \{1, \ldots, T_p\}$, the decay factor is calculated based on its temporal distance from the incident's occurrence:

$$\omega_\tau = \exp\left(-\frac{\tau^2}{2\sigma_t^2} \right), \tag{14}$$

where σ_t is a hyperparameter that controls the temporal decay rate. The decay factors for all time steps are combined into a vector $\boldsymbol{\omega} \in \mathbb{R}^{T_p}$. The initial incident context $\boldsymbol{C}_{\text{init}}$ is then linearly transformed and modulated by this decay vector via broadcast multiplication, yielding the final Temporal Incident Impact tensor $\boldsymbol{C}_{\text{temp}}$:

$$\boldsymbol{C}_{\text{temp}} = \boldsymbol{\omega} \otimes (\boldsymbol{C}_{\text{init}} \boldsymbol{W}_c), \tag{15}$$

where $\otimes$ denotes broadcasted multiplication.

Finally, the temporal incident impact tensor $\boldsymbol{C}_{\text{temp}}$ is integrated with the base forecast from the spatio-temporal backbone, $\boldsymbol{H}_{\text{pred}}$, to generate the final prediction tensor $\hat{\boldsymbol{Y}}$:

$$\hat{\boldsymbol{Y}} = g_{\text{out}}(\boldsymbol{H}_{\text{pred}} + \boldsymbol{C}_{\text{temp}}), \tag{16}$$

where g_{out} is a multi-layer perceptron serving as the prediction head.

4 Experiments

In this section, we experimentally validate the effectiveness of the proposed IGSTGNN framework. Following an introduction to the experimental setup (datasets, baselines, and metrics), we benchmark IGSTGNN against state-of-the-art models. The generalizability and design superiority of the core *ICSF* and *TIID* modules are then verified, followed by a detailed ablation study to assess the contribution of each component.

4.1 Experimental Settings

This subsection details our experimental setup, encompassing four key aspects: the datasets, the baseline models, the implementation details, and the metrics.

4.1.1 Datasets. Our experimental data are sourced from the large-scale XTraffic [14] benchmark dataset, which contains comprehensive traffic flow and incident records from California for the year 2023. To focus on freeway mainline traffic dynamics, we exclusively select data from mainline sensors. All data is aggregated into 5-minute time windows. We constructed three distinct sub-datasets for our evaluation, corresponding to three different regions in California: Alameda, Contra Costa, and Orange. Each dataset is partitioned into training, validation, and testing sets with a 70%/15%/15% ratio. Our task is formulated to predict the traffic conditions for the next 1 hour (12 steps) using the historical sequence from the past 1 hour (12 steps). Detailed statistics of the datasets are presented in Table 1.

Feature Details. To fully utilize the incident context, we extracted rich attributes from the raw logs. The specific **Sensor Meta-features** (e.g., road type, speed limit) and **Incident Information Features** (e.g., type, description, location) used in our model are detailed in Table 2 and Table 3, respectively.

Table 1: Statistics of the experimental datasets.

Dataset	# Nodes	# Edges	# Incidents
Alameda	521	13,828	14,687
Contra Costa	496	13,339	5,587
Orange	990	29,142	18,700

Table 2: Description of Sensor Meta-features.

Symbol	Feature Name	Type	Example
s_1	Type	String	Mainline
s_2	Surface	String	Asphalt
s_3	Roadway Use	String	Commercial
s_4	Lane Width	Float	3.7
s_5	Design Speed Limit	Integer	105
s_6, s_7	Latitude, Longitude	Float	34.0522, -118.2437
s_8	Freeway Name	String	I-405
s_9	Absolute Postmile	Float	23.15

Table 3: Description of Incident Information Features.

Symbol	Feature Name	Type	Example
e_1	Relative Position	Integer	11
e_2	Description	String	Traffic Collision
e_3	Type	String	Accident
e_4	Holiday	Cat.	0
e_5, e_6	Latitude, Longitude	Float	34.0592, -118.4452
e_7	Absolute Postmile	Float	25.50
e_8	Freeway Name	String	I-405

4.1.2 Baselines on Benchmarks. We compare our proposed IGSTGNN with a series of state-of-the-art spatio-temporal forecasting models. To ensure a fair comparison, our forecasting experiments were implemented within the same software framework employed by LargeST [25], and we further adhered to the identical experimental settings outlined within that work. The selected baselines include classic methods (HL [24], LSTM [18]) and a variety of spatio-temporal graph models. The latter can be further categorized by their temporal modeling approach into: **RNN-based** (DCRNN [23], AGCRN [6]), **CNN-based** (STGCN [41], GWNET [36]), **attention-based** (ASTGCN [15], STTN [38]), and recent **dynamic graph-based** approaches (DSTAGNN [21], DGCRN [22], D^{2}STGNN [30], BiST [26]).

4.1.3 Implementation Details. For our proposed IGSTGNN model, we use the Adam optimizer with an initial learning rate of 0.002 and an early stopping strategy (patience of 20 epochs) employed to prevent overfitting. The batch size was set to 48 for the Alameda and Contra Costa datasets, and adjusted to 24 for the larger Orange dataset. Key architectural hyperparameters were configured as follows: in the feature encoding stage, we construct 8-dimensional and 32-dimensional embeddings for the 6 incident types and 30 incident descriptions, respectively; the ST modeling module is composed of 5 decoupling blocks, where the dimension for both adaptive node and timestamp embeddings is set to 12; and in the *TIID* module, the temporal decay parameter σ_t is set to 1.0. All experiments were conducted on a server equipped with an Intel Xeon Gold 6138 CPU and an NVIDIA Tesla V100 (32GB) GPU[1].

4.1.4 Evaluation Metrics. Model performance is evaluated using three standard metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE), where lower values indicate better performance.

4.2 Performance Comparison with Baselines

We compare our proposed IGSTGNN model against all baselines on the three datasets to evaluate its overall effectiveness. The detailed quantitative results are summarized in Table 4, which reports the performance over three prediction horizons (15, 30, and 60 minutes ahead) as well as the average performance.

The results clearly demonstrate the superiority of our proposed IGSTGNN framework, which consistently outperforms all state-of-the-art baselines across all datasets and prediction horizons. For instance, on the Alameda dataset, IGSTGNN achieves an average MAE that is **5.65%** lower than the second-best model. This consistent advantage strongly validates the effectiveness of our incident-aware design. Key observations are as follows:

Robustness in Long-Term Forecasting. IGSTGNN consistently maintains its leading position in the most challenging long-term forecasting tasks. For example, on the Orange dataset, IGSTGNN still maintains a significant 4.1% advantage in MAE over the strong second-best model, STTN, at Horizon 12. This directly validates the effectiveness of our *TIID* module in capturing the dynamics of traffic recovery. This robustness is particularly important for real-world deployment, where operators rely on stable forecasts under rare but severe incidents.

[1]Code is available at https://github.com/fanlixiang/IGSTGNN.

Table 4: Performance comparison on the Alameda, Contra Costa, and Orange datasets. The best results are in bold and the second-best results are underlined.

Data	Method	Horizon 3			Horizon 6			Horizon 12			Average		
		MAE	RMSE	MAPE(%)	MAE	RMSE	MAPE(%)	MAE	RMSE	MAPE(%)	MAE	RMSE	MAPE(%)
Alameda	HL	15.60	26.44	18.62	18.48	31.50	21.61	24.86	41.63	26.98	19.06	32.24	21.72
	LSTM	13.88	23.15	17.34	15.86	26.86	20.00	19.79	32.98	22.47	16.04	26.82	19.83
	DCRNN	13.63	23.22	17.90	15.68	27.01	19.83	19.53	32.90	22.39	15.78	26.83	19.49
	AGCRN	13.07	22.01	17.81	14.70	25.01	19.46	16.81	27.79	22.47	14.34	24.03	19.78
	STGCN	15.73	25.30	22.26	17.68	28.88	28.14	21.31	35.43	29.70	17.79	29.09	25.65
	GWNET	14.87	24.48	18.93	17.69	29.49	21.45	22.51	37.29	31.44	17.76	29.49	22.77
	ASTGCN	14.60	24.69	17.00	16.79	28.86	21.26	21.86	36.66	30.79	17.16	29.08	22.08
	STTN	13.39	22.07	18.00	14.89	24.90	20.20	17.86	29.36	22.25	15.08	24.98	20.16
	DSTAGNN	12.48	21.25	14.74	13.53	23.13	15.62	15.24	25.27	30.10	13.45	22.63	19.28
	DGCRN	14.18	23.77	20.24	16.79	28.72	24.17	24.44	39.97	40.04	17.76	29.71	26.81
	D^{2}STGNN	13.19	22.41	15.40	14.64	24.90	16.65	17.09	28.39	19.91	14.62	24.67	17.09
	BiST	13.33	23.05	14.71	14.28	24.56	17.42	16.60	27.98	19.30	14.27	24.17	16.60
	IGSTGNN	**11.80**	**20.08**	**14.33**	**12.64**	**21.70**	**15.22**	**14.21**	**24.37**	**16.97**	**12.69**	**21.73**	**15.43**
Contra Costa	HL	16.47	27.85	16.58	18.64	31.28	20.65	23.44	37.52	24.65	19.15	31.71	19.95
	LSTM	14.78	24.95	15.63	16.89	29.02	19.65	20.04	32.74	19.73	16.88	28.31	17.74
	DCRNN	14.42	24.81	14.95	16.00	28.06	18.37	19.25	31.90	18.87	16.17	27.67	16.50
	AGCRN	14.65	24.36	14.96	15.91	27.06	18.78	18.41	29.38	19.18	15.98	26.51	16.94
	STGCN	16.39	24.66	**13.85**	20.75	30.82	**14.93**	23.12	33.77	19.85	19.71	29.35	**15.46**
	GWNET	14.69	24.67	17.16	15.93	27.78	20.16	19.16	30.89	19.21	16.46	27.50	18.24
	ASTGCN	14.94	25.38	15.75	17.15	29.25	22.72	20.44	33.00	27.42	17.31	28.68	20.93
	STTN	14.23	23.95	15.48	15.32	26.14	19.00	18.32	29.06	19.95	15.59	26.02	17.19
	DSTAGNN	14.35	24.51	14.75	15.48	26.32	19.02	16.56	27.19	**16.83**	15.28	25.74	16.65
	DGCRN	15.62	25.32	18.44	19.38	30.73	29.05	29.01	43.17	42.05	20.39	31.70	28.00
	D^{2}STGNN	12.91	21.76	17.26	14.03	23.47	20.45	16.01	25.70	23.89	14.09	23.31	20.19
	BiST	13.31	23.29	14.57	14.67	25.88	19.16	16.59	27.12	18.04	14.62	25.13	16.18
	IGSTGNN	**12.50**	**21.37**	14.14	**13.44**	**22.83**	16.01	**14.81**	**24.13**	17.95	**13.43**	**22.50**	15.84
Orange	HL	16.79	28.75	17.35	20.52	34.87	20.56	25.76	42.49	24.77	20.51	34.59	20.34
	LSTM	14.47	24.70	15.43	16.94	29.53	17.63	20.66	34.91	19.61	17.02	29.24	17.29
	DCRNN	14.07	24.22	14.24	16.11	27.73	15.92	19.11	32.57	18.00	16.11	27.68	15.95
	AGCRN	14.48	23.44	20.74	16.48	27.73	19.84	18.34	30.79	22.61	16.20	27.19	20.79
	STGCN	15.82	24.81	20.07	18.22	29.61	21.35	19.76	32.31	21.22	17.79	28.99	20.82
	GWNET	14.69	24.70	16.33	17.14	29.77	19.07	19.95	33.81	21.12	16.99	29.01	18.40
	ASTGCN	14.71	25.50	15.65	17.66	31.10	18.63	21.83	37.06	24.12	17.56	30.49	18.89
	STTN	13.09	22.54	14.54	13.91	24.01	15.61	14.96	25.74	16.60	13.82	23.80	15.39
	DSTAGNN	13.69	22.97	19.44	14.36	25.03	16.02	15.69	27.03	**16.06**	14.61	25.03	16.51
	DGCRN	13.99	23.53	17.75	16.00	27.01	22.81	19.89	31.77	33.36	16.14	26.84	23.25
	D^{2}STGNN	13.11	22.83	15.68	13.93	24.27	17.28	15.31	26.52	18.02	13.92	24.21	16.83
	BiST	13.04	22.68	15.76	14.65	25.37	17.37	16.75	29.16	19.72	14.56	25.39	17.23
	IGSTGNN	**12.28**	**21.95**	**13.82**	**13.16**	**23.45**	**15.16**	**14.35**	**25.48**	16.10	**13.13**	**23.35**	**14.80**

Adaptability to Spatial Heterogeneity. IGSTGNN maintains its leading performance despite the varying network structures and incident distributions across the three datasets. This suggests that our *ICSF* module can effectively handle the spatial heterogeneity of incident impacts by generating a customized incident impact representation for each node.

4.3 Effectiveness of Proposed Modules

To validate the generalizability of our proposed *ICSF* and *TIID* modules as "plug-and-play" components, we conducted a module effectiveness study. To ensure a comprehensive evaluation, we selected several state-of-the-art STGNNs from our baselines that represent distinct architectural paradigms: an RNN-based model (AGCRN), a CNN-based model (GWNET), a Transformer-based model (STTN), and a hybrid model employing a spatio-temporal decoupling framework (D^{2}STGNN). We integrated the *ICSF* and *TIID* modules with these models in three settings: *ICSF* only, *TIID* only, and both modules combined, evaluating their performance changes across different datasets.

The results are visualized in Figure 3. It is clearly observable that while integrating either the *ICSF* or *TIID* module alone brings consistent performance gains to all tested baselines, combining

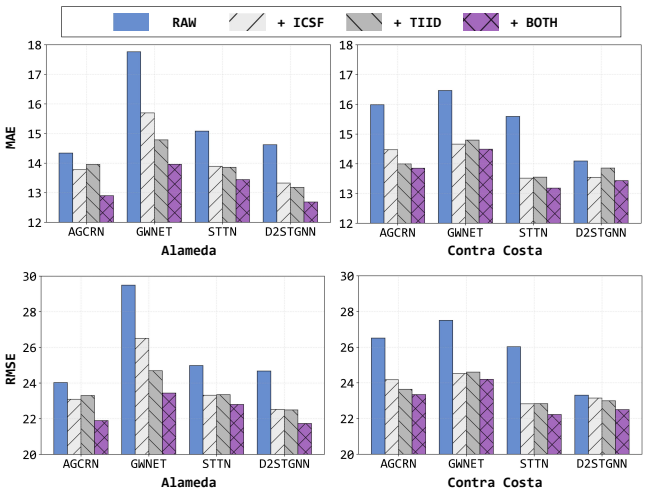


Figure 3: Effectiveness of the *ICSF* and *TIID* modules. The figure compares the performance (MAE and RMSE) of baseline models in their original form (Raw) versus after integrating the modules separately (*ICSF, TIID*) and in combination (BOTH) on the Alameda and Contra Costa datasets.

both modules achieves the most significant improvements, demonstrating a strong synergistic effect. For instance, on the Alameda dataset, integrating the *ICSF* and *TIID* modules with the D^{2}STGNN model separately yields MAE improvements of **8.85%** and **9.85%**, respectively. When both modules were enabled concurrently, the performance improvement soared to **13.23%**, proving their functional complementarity. This trend is highly consistent across diverse model architectures and datasets. These findings strongly demonstrate that our proposed *ICSF* and *TIID* modules are not only effective on their own but also work synergistically as general, "plug-and-play" enhancement components. They can empower other spatio-temporal forecasting models with the crucial capabilities of contextual fusion and temporal decay modeling required for handling traffic incidents.

4.4 Superiority of the *ICSF* Module

Finally, to demonstrate the superiority of our designed *ICSF* module compared to simpler fusion methods, we conduct a comparative experiment. In this experiment, we replace the *ICSF* module within the full IGSTGNN framework with two alternatives: The first is **MLP**, which simply concatenates the embedding vectors of all information and fuses them through an MLP. The second is **Iterative Message Passing (IMP)**, which employs a more complex, message-passing-like mechanism where incident and node representations undergo multiple rounds of interactive updates.

As shown in Figure 4, our proposed context-aware attention-based *ICSF* module significantly outperforms both the MLP and IMP alternatives in terms of average MAE and RMSE on both datasets. Specifically, on the Alameda dataset, the MAE of *ICSF* is **10.3%** lower than that of MLP and **12.5%** lower than that of IMP. This result validates our design choice: simple feature concatenation

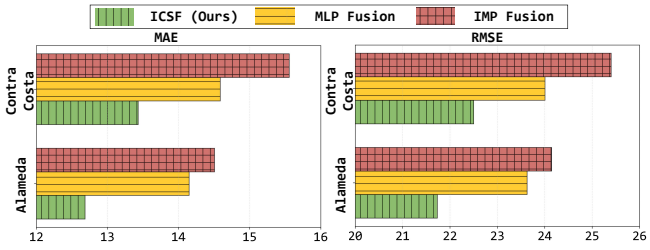


Figure 4: Performance comparison (in terms of average MAE and RMSE) of our *ICSF* module against MLP and IMP fusion methods on the Alameda and Contra Costa datasets.

(MLP) is insufficient to capture the complex non-linear relationships between incidents, sensors, and traffic states, while overly complex interactions (IMP) may be difficult to optimize. In contrast, the attention mechanism in *ICSF* strikes a better balance between capturing critical information and modeling complex relationships, thereby achieving superior performance.

4.5 Ablation Study

To deeply investigate the effectiveness of our model design, we conduct a series of detailed ablation studies. We analyze the contributions from two perspectives: (1) **Module Ablation**, where we remove the *ICSF* or *TIID* modules respectively, and (2) **Feature Ablation**, where we remove sensor meta-features, incident intrinsic attributes, or distance information.

Table 5: Ablation study of IGSTGNN and its variants on Alameda, Contra Costa, and Orange datasets. All metrics are averaged over the 12-step prediction horizon.

Data	Method	Average		
		MAE	RMSE	MAPE(%)
Alameda	w/o *ICSF*	13.23	22.68	17.27
	w/o *TIID*	13.23	22.64	**15.13**
	w/o S (sensor attr.)	13.21	22.61	15.44
	w/o D (distance)	13.20	22.61	18.44
	w/o I (incident attr.)	13.09	22.48	15.96
	IGSTGNN (Ours)	**12.69**	**21.73**	15.43
Contra Costa	w/o *ICSF*	13.52	23.25	16.76
	w/o *TIID*	13.54	22.83	17.35
	w/o S (sensor attr.)	13.52	23.08	16.14
	w/o D (distance)	13.55	23.24	17.16
	w/o I (incident attr.)	13.59	22.99	**15.80**
	IGSTGNN (Ours)	**13.43**	**22.50**	15.84
Orange	w/o *ICSF*	13.38	23.60	14.84
	w/o *TIID*	13.32	23.57	14.82
	w/o S (sensor attr.)	13.45	23.68	**14.53**
	w/o D (distance)	13.60	23.72	17.50
	w/o I (incident attr.)	13.59	23.66	15.55
	IGSTGNN (Ours)	**13.13**	**23.35**	14.80

Table 5 presents the quantitative results of the ablation study across three datasets. Specifically, we removed each component or

feature input to isolate its contribution. The results consistently show that the full IGSTGNN achieves the best MAE/RMSE across all three datasets, while MAPE is largely comparable with only minor variations in a few cases, supporting the overall effectiveness of the proposed framework. We draw three key observations:

First, both the *ICSF* and *TIID* modules contribute consistently to performance. Removing either module increases MAE and RMSE across all datasets, though the magnitude varies by dataset. For instance, on Alameda, removing *ICSF* or *TIID* increases the MAE to 13.23 (a 4.2% degradation compared to the full model), indicating that modeling both the initial spatial heterogeneity and the temporal decay of incident impacts is important.

Second, incorporating **incident attributes (I)** provides complementary gains. The variant "w/o I" (removing incident type and description inputs) shows a degradation (e.g., MAE increases from 13.13 to 13.59 on Orange), suggesting that incident semantics offer additional discriminative cues beyond spatial proximity and sensor context.

Finally, spatial context features, including **Distance (D)** and **Sensor Attributes (S)**, also play an important role. In particular, removing distance information leads to a notable degradation (e.g., MAPE rises to 17.50 on Orange), confirming that precise spatial anchoring and road network context are crucial for modeling how incident impacts propagate through the traffic graph.

5 Related Work

This section reviews the relevant literature from two perspectives: the mainstream models for spatio-temporal traffic forecasting and related works about the impact of external incidents on traffic.

5.1 Spatio-Temporal Traffic Forecasting

Spatio-temporal traffic forecasting has evolved from traditional statistical methods (e.g., ARIMA) and early deep learning models (e.g., LSTM), which have limitations in capturing complex spatio-temporal dependencies [8, 29, 41, 43]. Currently, the dominant paradigm involves combining Graph Neural Networks with various temporal modeling components. The core idea is to abstract the traffic network as a graph, utilizing Graph Neural Networks (GNNs) to capture spatial dependencies while combining them with Recurrent Neural Networks (RNNs), Convolutional Neural Networks (CNNs), or attention mechanisms to model temporal dynamics [7, 8, 10, 11, 16, 19, 27, 31, 35, 41].

Recent works have further explored decoupling complex spatio-temporal patterns and adaptively learning the graph structure [19, 31]. Based on their temporal modeling approach, these advanced models can be categorized into: **RNN-based** (DCRNN [23], AGCRN [6]), **CNN-based** (STGCN [41], GWNET [36]), **attention-based** (ASTGCN [15], STTN [38]), and recent **dynamic graph-based** approaches (DSTAGNN [21], DGCRN [22], D^{2}STGNN [30], BiST [26]). Despite the great success of these graph-based deep learning models in learning from historical traffic data, their core design is oriented towards regular, periodic traffic patterns, which limits their ability to model the atypical dynamics induced by external, abrupt incidents.

5.2 The Incident Impact in Traffic Analysis

Incorporating external information has become a key research direction for improving traffic forecasting accuracy, especially when purely historical signals fail to explain abrupt state transitions [1, 2, 46–48]. Among various external sources, traffic incidents (e.g., accidents, road works, and control measures) have attracted significant attention due to their strong association with congestion and sudden changes in traffic states [1, 2, 48]. Moreover, the multi-dimensional attributes of incidents—such as type, location, and duration—provide rich, quantifiable features for modeling traffic flow variations [46–48].

More recently, specialized incident-aware models have been proposed to better handle the irregularity and sparsity of incident patterns. For example, DIGC-Net introduces a two-stage critical incident discovery and impact representation learning pipeline, which identifies critical incidents and aggregates sequential incident information within a time window before fusing it with spatio-temporal predictors [37]. In parallel, broader event-aware frameworks (beyond road incidents) have been explored for mobility forecasting under societal events such as holidays, severe weather, and epidemics; EAST-Net leverages memory-augmented dynamic parameter generation to enhance adaptivity under unprecedented events [34].

Despite these advances, incident effects are often incorporated through implicit feature fusion (e.g., concatenation/attention/sequence aggregation) rather than being explicitly formulated as a spatio-temporal propagation and attenuation process on the traffic network. As a result, existing methods may still be limited in capturing (i) heterogeneous spatial spread of localized disruptions and (ii) the continuous temporal evolution and decay of incident impacts.

6 Conclusion

In this paper, we proposed IGSTGNN, a novel framework designed to address the challenges of forecasting in incident-affected scenarios. To systematically model an incident's influence, the framework utilized two core components: an Incident-Context Spatial Fusion (*ICSF*) module to capture the non-uniform spatial impact, and a Temporal Incident Impact Decay (*TIID*) module to model the dynamic temporal evolution of the impact. Comprehensive experiments on multiple large-scale, real-world datasets demonstrated that the performance of IGSTGNN was significantly superior to state-of-the-art baseline models. Our work provides an effective and systematic solution for incident-guided spatio-temporal forecasting. To support reproducibility, we release the dataset to facilitate future research and fair comparisons. Moreover, the proposed *ICSF* and *TIID* modules can be integrated as plug-and-play components, consistently improving several STGNN backbones. Future work could explore extending the framework to incorporate a broader range of external disturbances, and leveraging foundation models or agentic pipelines to extract and normalize incident semantics.

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References

[1] A. Abadi, Tooraj Rajabioun, and Petros A. Ioannou. 2015. Traffic Flow Prediction for Road Transportation Networks With Limited Traffic Data. *IEEE Transactions on Intelligent Transportation Systems* 16 (2015), 653–662. https://doi.org/10.1109/TITS.2014.2337238

[2] Shams Forruque Ahmed, Sweety Angela Kuldeep, Sabiha Jannat Rafa, Javeria Fazal, Mahfara Hoque, Gang Liu, and Amir H. Gandomi. 2024. Enhancement of traffic forecasting through graph neural network-based information fusion techniques. *Inf. Fusion* 110 (2024), 102466. https://doi.org/10.1016/j.inffus.2024.102466

[3] Mansoor G. Al-Thani, Ziyu Sheng, Yuting Cao, and Yin Yang. 2024. Traffic Transformer: Transformer-based framework for temporal traffic accident prediction. *AIMS Mathematics* 9, 5 (2024), 12610–12629. https://doi.org/10.3934/math.2024617

[4] Ahmad Ali, Yanmin Zhu, and M. Zakarya. 2021. Exploiting dynamic spatio-temporal graph convolutional neural networks for citywide traffic flows prediction. *Neural networks : the official journal of the International Neural Network Society* 145 (2021), 233–247. https://doi.org/10.1016/j.neunet.2021.10.021

[5] Jiyao An, Liang Guo, Wei Liu, Zhiqiang Fu, Ping Ren, Xinzhi Liu, and Tao Li. 2021. IGAGCN: Information geometry and attention-based spatiotemporal graph convolutional networks for traffic flow prediction. *Neural Networks* 143 (2021), 355–367.

[6] Lei Bai, Lina Yao, Can Li, Xianzhi Wang, and Can Wang. 2020. Adaptive graph convolutional recurrent network for traffic forecasting. *Advances in neural information processing systems* 33 (2020), 17804–17815.

[7] Razvan-Gabriel Cirstea, Bin Yang, Chenjuan Guo, Tung Kieu, and Shirui Pan. 2022. Towards Spatio- Temporal Aware Traffic Time Series Forecasting. In *38th IEEE International Conference on Data Engineering, ICDE 2022, Kuala Lumpur, Malaysia, May 9-12, 2022*. IEEE, Kuala Lumpur, Malaysia, 2900–2913. https://doi.org/10.1109/ICDE53745.2022.00262

[8] Fei Dai, Penggui Huang, Xiaolong Xu, Lianyong Qi, and M. Khosravi. 2020. Spatio-Temporal Deep Learning Framework for Traffic Speed Forecasting in IoT. *IEEE Internet of Things Magazine* 3 (2020), 66–69. https://doi.org/10.1109/IOTM.0001.2000031

[9] Bowen Du, Hao Peng, Senzhang Wang, Md Zakirul Alam Bhuiyan, Lihong Wang, Qiran Gong, Lin Liu, and Jing Li. 2019. Deep irregular convolutional residual LSTM for urban traffic passenger flows prediction. *IEEE Transactions on Intelligent Transportation Systems* 21, 3 (2019), 972–985.

[10] Shengdong Du, Tao Yang, Fei Teng, Junbo Zhang, Tianrui Li, and Yu Zheng. 2024. Multi-scale feature enhanced spatio-temporal learning for traffic flow forecasting. *Knowl. Based Syst.* 294 (2024), 111787. https://doi.org/10.1016/j.knosys.2024.111787

[11] Yuchen Fang, Fang Zhao, Yanjun Qin, Haiyong Luo, and Chenxing Wang. 2022. Learning All Dynamics: Traffic Forecasting via Locality-Aware Spatio-Temporal Joint Transformer. *IEEE Transactions on Intelligent Transportation Systems* 23 (2022), 23433–23446. https://doi.org/10.1109/TITS.2022.3197640

[12] Siyuan Feng, Shuqing Wei, Junbo Zhang, Yexin Li, Jintao Ke, Gaode Chen, Yu Zheng, and Hai Yang. 2023. A macro–micro spatio-temporal neural network for traffic prediction. *Transportation Research Part C: Emerging Technologies* 156 (2023), 104331. https://doi.org/10.1016/j.trc.2023.104331

[13] Banishree Ghosh and J. Dauwels. 2021. Comparison of different Bayesian methods for estimating error bars with incident duration prediction. *Journal of Intelligent Transportation Systems* 26 (2021), 420 – 431. https://doi.org/10.1080/15472450.2021.1894936

[14] Xiaochuan Gou, Ziyue Li, Tian Lan, Junpeng Lin, Zhishuai Li, Bingyu Zhao, Chen Zhang, Di Wang, and Xiangliang Zhang. 2024. XTraffic: A Dataset Where Traffic Meets Incidents with Explainability and More. arXiv:2407.11477 [cs.LG] https://arxiv.org/abs/2407.11477

[15] Shengnan Guo, Youfang Lin, Ning Feng, Chao Song, and Huaiyu Wan. 2019. Attention Based Spatial-Temporal Graph Convolutional Networks for Traffic Flow Forecasting. In *The Thirty-Third AAAI Conference on Artificial Intelligence, AAAI 2019, The Thirty-First Innovative Applications of Artificial Intelligence Conference, IAAI 2019, The Ninth AAAI Symposium on Educational Advances in Artificial Intelligence, EAAI 2019, Honolulu, Hawaii, USA, January 27 - February 1, 2019*. AAAI Press, Honolulu, HI, USA, 922–929. https://doi.org/10.1609/AAAI.V33I01.3301922

[16] Liangzhe Han, Bowen Du, Leilei Sun, Yanjie Fu, Yisheng Lv, and Hui Xiong. 2021. Dynamic and Multi-faceted Spatio-temporal Deep Learning for Traffic Speed Forecasting. In *KDD '21: The 27th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, Virtual Event, Singapore, August 14-18, 2021*, Feida Zhu, Beng Chin Ooi, and Chunyan Miao (Eds.). ACM, Singapore, 547–555. https://doi.org/10.1145/3447548.3467275

[17] Xiaolin Han, Tobias Grubenmann, Reynold Cheng, Sze Chun Wong, Xiaodong Li, and Wenya Sun. 2020. Traffic Incident Detection: A Trajectory-based Approach. In *36th IEEE International Conference on Data Engineering, ICDE 2020, Dallas, TX, USA, April 20-24, 2020*. IEEE, Dallas, TX, USA, 1866–1869. https://doi.org/10.1109/ICDE48307.2020.00190

[18] Sepp Hochreiter and Jürgen Schmidhuber. 1997. Long short-term memory. *Neural computation* 9, 8 (1997), 1735–1780.

[19] Renhe Jiang, Zhaonan Wang, Jiawei Yong, Puneet Jeph, Quanjun Chen, Yasumasa Kobayashi, Xuan Song, Shintaro Fukushima, and Toyotaro Suzumura. 2023. Spatio-Temporal Meta-Graph Learning for Traffic Forecasting. In *Thirty-Seventh AAAI Conference on Artificial Intelligence, AAAI 2023, Thirty-Fifth Conference on Innovative Applications of Artificial Intelligence, IAAI 2023, Thirteenth Symposium on Educational Advances in Artificial Intelligence, EAAI 2023, Washington, DC, USA, February 7-14, 2023*, Brian Williams, Yiling Chen, and Jennifer Neville (Eds.). AAAI Press, Washington, DC, USA, 8078–8086. https://doi.org/10.1609/AAAI.V37I7.25976

[20] Wei Ju, Yusheng Zhao, Yifang Qin, Siyu Yi, Jingyang Yuan, Zhiping Xiao, Xiao Luo, Xiting Yan, and Ming Zhang. 2024. COOL: A Conjoint Perspective on Spatio-Temporal Graph Neural Network for Traffic Forecasting. *Inf. Fusion* 107 (2024), 102341. https://doi.org/10.48550/arXiv.2403.01091

[21] Shiyong Lan, Yitong Ma, Weikang Huang, Wenwu Wang, Hongyu Yang, and Pyang Li. 2022. DSTAGNN: Dynamic Spatial-Temporal Aware Graph Neural Network for Traffic Flow Forecasting. In *International Conference on Machine Learning, ICML 2022, 17-23 July 2022, Baltimore, Maryland, USA (Proceedings of Machine Learning Research, Vol. 162)*, Kamalika Chaudhuri, Stefanie Jegelka, Le Song, Csaba Szepesvári, Gang Niu, and Sivan Sabato (Eds.). PMLR, Baltimore, Maryland, USA, 11906–11917. https://proceedings.mlr.press/v162/lan22a.html

[22] Fuxian Li, Jie Feng, Huan Yan, Guangyin Jin, Fan Yang, Funing Sun, Depeng Jin, and Yong Li. 2023. Dynamic graph convolutional recurrent network for traffic prediction: Benchmark and solution. *ACM Transactions on Knowledge Discovery from Data* 17, 1 (2023), 1–21.

[23] Yaguang Li, Rose Yu, Cyrus Shahabi, and Yan Liu. 2018. Diffusion Convolutional Recurrent Neural Network: Data-Driven Traffic Forecasting. In *6th International Conference on Learning Representations, ICLR 2018, Vancouver, BC, Canada, April 30 - May 3, 2018, Conference Track Proceedings*. OpenReview.net, Vancouver, BC, Canada. https://openreview.net/forum?id=SJiHXGWAZ

[24] Yuxuan Liang, Kun Ouyang, Yiwei Wang, Ye Liu, Junbo Zhang, Yu Zheng, and David S. Rosenblum. 2020. Revisiting Convolutional Neural Networks for Citywide Crowd Flow Analytics. In *Machine Learning and Knowledge Discovery in Databases - European Conference, ECML PKDD 2020, Ghent, Belgium, September 14-18, 2020, Proceedings, Part I (Lecture Notes in Computer Science, Vol. 12457)*, Frank Hutter, Kristian Kersting, Jefrey Lijffijt, and Isabel Valera (Eds.). Springer, Ghent, Belgium, 578–594. https://doi.org/10.1007/978-3-030-67658-2_33

[25] Xu Liu, Yutong Xia, Yuxuan Liang, Junfeng Hu, Yiwei Wang, Lei Bai, Chao Huang, Zhenguang Liu, Bryan Hooi, and Roger Zimmermann. 2023. Largest: A benchmark dataset for large-scale traffic forecasting. *Advances in Neural Information Processing Systems* 36 (2023), 75354–75371.

[26] Jiaming Ma, Binwu Wang, Pengkun Wang, Zhengyang Zhou, Xu Wang, and Yang Wang. 2025. BiST: A Lightweight and Efficient Bi-directional Model for Spatiotemporal Prediction. *Proc. VLDB Endow.* 18, 6 (2025), 1663–1676. https://www.vldb.org/pvldb/vol18/p1663-wang.pdf

[27] Qiwei Ma, Wei Sun, Junbo Gao, Pengwei Ma, and Mengjie Shi. 2022. Spatio-temporal adaptive graph convolutional networks for traffic flow forecasting. *IET Intelligent Transport Systems* 17, 4 (Oct. 2022), 691–703. https://doi.org/10.1049/itr2.12296

[28] Boris Medina-Salgado, Eddy Sánchez-Delacruz, Maria del Pilar Pozos Parra, and J. Sierra. 2022. Urban traffic flow prediction techniques: A review. *Sustain. Comput. Informatics Syst.* 35 (2022), 100739. https://doi.org/10.1016/j.suscom.2022.100739

[29] Zheyi Pan, Yuxuan Liang, Weifeng Wang, Yong Yu, Yu Zheng, and Junbo Zhang. 2019. Urban Traffic Prediction from Spatio-Temporal Data Using Deep Meta Learning. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining, KDD 2019, Anchorage, AK, USA, August 4-8, 2019*, Ankur Teredesai, Vipin Kumar, Ying Li, Rómer Rosales, Evimaria Terzi, and George Karypis (Eds.). ACM, Anchorage, AK, USA, 1720–1730. https://doi.org/10.1145/3292500.3330884

[30] Zezhi Shao, Zhao Zhang, Wei Wei, Fei Wang, Yongjun Xu, Xin Cao, and Christian S. Jensen. 2022. Decoupled Dynamic Spatial-Temporal Graph Neural Network for Traffic Forecasting. *Proc. VLDB Endow.* 15, 11 (2022), 2733–2746. https://doi.org/10.14778/3551793.3551827

[31] Xuxiang Ta, Zihan Liu, Xiao Hu, Le Yu, Leilei Sun, and Bowen Du. 2022. Adaptive Spatio-temporal Graph Neural Network for traffic forecasting. *Knowl. Based Syst.* 242 (2022), 108199. https://doi.org/10.1016/j.knosys.2022.108199

[32] Thanh Tran, Dan He, Jiwon Kim, and Mark Hickman. 2023. MSGNN: A Multi-structured Graph Neural Network model for real-time incident prediction in large traffic networks. *Transportation research part C: emerging technologies* 156 (2023), 104354.

[33] Yi Wang, Changfeng Jing, Shisuo Xu, and Tao Guo. 2022. Attention based spatiotemporal graph attention networks for traffic flow forecasting. *Inf. Sci.* 607 (2022), 869–883. https://doi.org/10.1016/j.ins.2022.05.127

[34] Zhaonan Wang, Renhe Jiang, Hao Xue, Flora D. Salim, Xuan Song, and Ryosuke Shibasaki. 2022. Event-Aware Multimodal Mobility Nowcasting. In *Thirty-Sixth AAAI Conference on Artificial Intelligence, AAAI 2022, Thirty-Fourth Conference on Innovative Applications of Artificial Intelligence, IAAI 2022, The Twelveth Symposium on Educational Advances in Artificial Intelligence, EAAI 2022*

Virtual Event, February 22 - March 1, 2022. AAAI Press, 4228–4236. https://doi.org/10.1609/AAAI.V36I4.20342

[35] Cong Wu, Hui Ding, Zhongwang Fu, and Ning Sun. 2024. Multi-Scale Spatio-Temporal Attention Networks for Network-Scale Traffic Learning and Forecasting. *Sensors* 24, 17 (2024), 5543. https://doi.org/10.3390/S24175543

[36] Zonghan Wu, Shirui Pan, Guodong Long, Jing Jiang, and Chengqi Zhang. 2019. Graph WaveNet for Deep Spatial-Temporal Graph Modeling. In *Proceedings of the Twenty-Eighth International Joint Conference on Artificial Intelligence, IJCAI 2019, Macao, China, August 10-16, 2019*, Sarit Kraus (Ed.). ijcai.org, Macao, China, 1907–1913. https://doi.org/10.24963/IJCAI.2019/264

[37] Qinge Xie, Tiancheng Guo, Yang Chen, Yu Xiao, Xin Wang, and Ben Y. Zhao. 2020. Deep Graph Convolutional Networks for Incident-Driven Traffic Speed Prediction. In *CIKM '20: The 29th ACM International Conference on Information and Knowledge Management, Virtual Event, Ireland, October 19-23, 2020*, Mathieu d'Aquin, Stefan Dietze, Claudia Hauff, Edward Curry, and Philippe Cudré-Mauroux (Eds.). ACM, 1665–1674. https://doi.org/10.1145/3340531.3411873

[38] Mingxing Xu, Wenrui Dai, Chunmiao Liu, Xing Gao, Weiyao Lin, Guo-Jun Qi, and Hongkai Xiong. 2021. Spatial-Temporal Transformer Networks for Traffic Flow Forecasting. arXiv:2001.02908 [eess.SP] https://arxiv.org/abs/2001.02908

[39] Zhijian Xu, Hao Wang, and Qiang Xu. 2025. Intervention-Aware Forecasting: Breaking Historical Limits from a System Perspective. arXiv:2405.13522 [cs.LG] https://arxiv.org/abs/2405.13522

[40] Yaqin Ye, Yue Xiao, Yuxuan Zhou, S. Li, Yuanfei Zang, and Yixuan Zhang. 2023. Dynamic multi-graph neural network for traffic flow prediction incorporating traffic accidents. *Expert Syst. Appl.* 234 (2023), 121101. https://doi.org/10.2139/ssrn.4382825

[41] Bing Yu, Haoteng Yin, and Zhanxing Zhu. 2018. Spatio-Temporal Graph Convolutional Networks: A Deep Learning Framework for Traffic Forecasting. In *Proceedings of the Twenty-Seventh International Joint Conference on Artificial Intelligence, IJCAI-18*. International Joint Conferences on Artificial Intelligence Organization, Stockholm, Sweden, 3634–3640. https://doi.org/10.24963/ijcai.2018/505

[42] Zhuoning Yuan, Xun Zhou, and Tianbao Yang. 2018. Hetero-ConvLSTM: A Deep Learning Approach to Traffic Accident Prediction on Heterogeneous Spatio-Temporal Data. In *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining, KDD 2018, London, UK, August 19-23, 2018*, Yike Guo and Faisal Farooq (Eds.). ACM, London, UK, 984–992. https://doi.org/10.1145/3219819.3219922

[43] Weibin Zhang, Yinghao Yu, Yong Qi, Feng Shu, and Yinhai Wang. 2019. Short-term traffic flow prediction based on spatio-temporal analysis and CNN deep learning. *Transportmetrica A: Transport Science* 15 (2019), 1688 – 1711. https://doi.org/10.1080/23249935.2019.1637966

[44] Xu Zhang, Shunjie Wen, Liang Yan, Jiangfan Feng, and Ying Xia. 2024. A hybrid-convolution spatial–temporal recurrent network for traffic flow prediction. *Comput. J.* 67, 1 (2024), 236–252.

[45] Ling Zhao, Yujiao Song, Chao Zhang, Yu Liu, Pu Wang, Tao Lin, Min Deng, and Haifeng Li. 2019. T-GCN: A temporal graph convolutional network for traffic prediction. *IEEE transactions on intelligent transportation systems* 21, 9 (2019), 3848–3858.

[46] Yi Zhou, Yihan Liu, Nianwen Ning, Li Wang, Zixing Zhang, Xiaozhi Gao, and Ning Lu. 2023. Integrating knowledge representation into traffic prediction: a spatial–temporal graph neural network with adaptive fusion features. *Complex & Intelligent Systems* 10 (2023), 2883–2900. https://doi.org/10.1007/s40747-023-01299-7

[47] Jiawei Zhu, Xing Han, Hanhan Deng, Chao Tao, Ling Zhao, Pu Wang, Tao Lin, and Haifeng Li. 2020. KST-GCN: A Knowledge-Driven Spatial-Temporal Graph Convolutional Network for Traffic Forecasting. *IEEE Transactions on Intelligent Transportation Systems* 23 (2020), 15055–15065. https://doi.org/10.1109/TITS.2021.3136287

[48] Jiawei Zhu, Qiongjie Wang, Chao Tao, Hanhan Deng, Ling Zhao, and Haifeng Li. 2020. AST-GCN: Attribute-Augmented Spatiotemporal Graph Convolutional Network for Traffic Forecasting. *IEEE Access* 9 (2020), 35973–35983. https://doi.org/10.1109/ACCESS.2021.3062114

A Appendix

A.1 Data Processing Details

The construction of the final dataset involved a multi-stage pipeline, ensuring precise alignment between traffic time series and incident logs:

- **Data Filtering and Selection:** We began by loading the raw sensor metadata and incident logs. To create focused and manageable datasets, we filtered this data to include only mainline sensors from three specific California counties (Alameda, Contra Costa, and Orange) for the year 2023.
- **Temporal Alignment:** The continuous traffic sensor readings were aggregated into discrete 5-minute time windows. Each incident record was then precisely aligned and mapped to a single 5-minute window based on its occurrence timestamp. This step is crucial for synchronizing the incident information with the traffic time series.
- **Geocoding and Spatial Relation Tensor Construction:** To compute spatial relationships, a unified coordinate system is required. While sensors had latitude and longitude, incidents were located by postmiles. We used the official **Caltrans Postmile Query Tool** to convert incident postmiles into precise latitude and longitude coordinates. With this information, we constructed the spatial relationship tensor (D), whose three dimensions for each incident-sensor pair represent their Gaussian-transformed Euclidean distance, Gaussian-transformed road network distance, and their binary upstream/downstream relationship.
- **Feature Engineering and Data Splitting:** Additional time-based features, such as the time of day and day of the week, were engineered from the timestamps. Finally, the complete time-ordered dataset for each region was chronologically split into training (70%), validation (15%), and testing (15%) sets.

A.2 Incident Taxonomy

To clarify the scope of external disturbances modeled in this study, we list the specific incident types and descriptions included in our dataset in Table 6. The dataset covers 6 major categories and 30 fine-grained descriptions.

Table 6: Taxonomy and Definitions of Incident Types.

Incident Type	Typical Examples	Key Definitions / Clarification
Hazard	Traffic Hazard, Debris	Impediments obstructing traffic flow (e.g., objects/animals).
Accident	1141 En Route, Collision	Vehicle collisions. **"1141"**: Injury accident requiring ambulance response.
Breakdown	Disabled Vehicle, Flat Tire	Vehicles stopped due to mechanical failures (non-collision).
Weather	Fog, Wind, Rain, Snow	Adverse atmospheric conditions affecting visibility/friction.
Other	Fire, Sigalert, Roadwork	Misc. events. **"Fire"**: Vehicle or roadside fire.
Police	Police Activity, Advisory	Law enforcement activities not directly involving collisions.

A.3 Supplementary Experimental Results

A.3.1 Performance by Incident Type. To evaluate the generalization capability of IGSTGNN across different disturbances, we analyzed the model's performance on subsets of test data filtered by incident type. As shown in Table 7, IGSTGNN consistently achieves low prediction errors across diverse categories, outperforming baselines even when they are enhanced with incident features.

Table 7: Performance comparison (MAE) across different incident types on the Alameda dataset.

Incident Type	IGSTGNN (Ours)	DSTAGNN+I	D^2STGNN+I
Hazard	**12.82**	13.02	13.37
Accident	**13.22**	13.52	14.17
Breakdown	**12.40**	13.01	12.94
Weather	**8.50**	9.36	10.56
Other	**14.25**	14.47	14.98
Police	**15.11**	15.62	16.22

A.3.2 ICSF Module Superiority Study. In this section, we compare *ICSF* with two representative baseline fusion approaches, which are detailed below.

MLP Fusion. This method serves as a simple fusion baseline. It simply **concatenates** all relevant feature vectors (e.g., incident and sensor representations) into a single, flattened vector. This combined vector is then processed by a Multi-Layer Perceptron (MLP) to produce a fused output, without explicitly modeling any structural or spatial relationships.

IMP Fusion (Iterative Message Passing). This is a more complex, graph-inspired approach. In this mechanism, incident and sensor nodes are treated as nodes in a bipartite graph. For a fixed number of iterations, they exchange information through a **message-passing** process, where each node updates its representation based on the aggregated messages from its neighbors in the previous round.

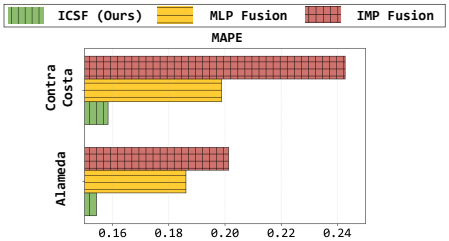


Figure 5: MAPE (%) results of the *ICSF* module superiority study.

The MAPE results, shown in Figure 5, demonstrate the superiority of our proposed *ICSF* module. On both datasets, *ICSF* achieves the lowest MAPE. This result validates our design choice: simple feature concatenation (MLP) is insufficient to capture complex spatial relationships, while an overly complex iterative method (IMP) may be difficult to optimize. In contrast, our *ICSF* module strikes a more effective balance between fusion effectiveness and model complexity.